

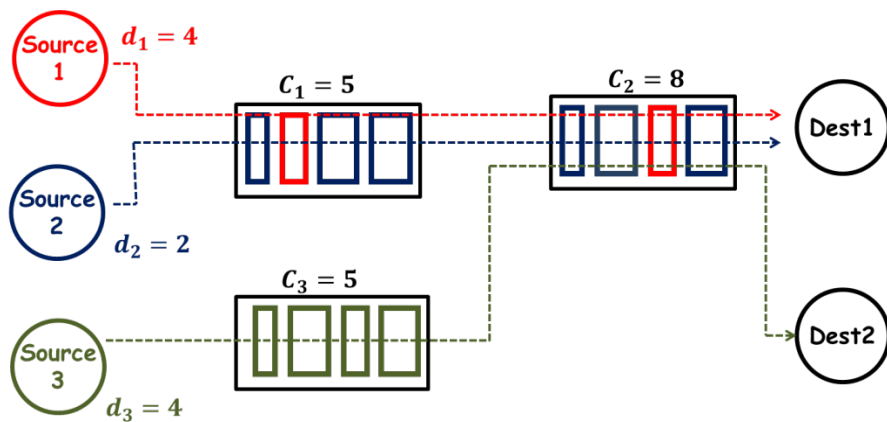
CS4226 Written Assignment

Problem 1

You flip a biased coin many times. Each time, you get a head with probability p and a tail with probability $1 - p$ independently. We define a random variable X to be the number of trials before you get the first head.

1. What is the distribution of X ? Write down the probability distribution function $F(x) = P\{X \leq x\}$?
2. What is the memoryless property? Explain it in words.
3. Show that the random variable X satisfies the memoryless property. State the memoryless property of X in a mathematical statement first, and then prove it.

Problem 2



Consider the above topology with three flows and three links. The three flows have demands $d_1 = 4$, $d_2 = 2$ and $d_3 = 4$. The three links have capacities $C_1 = 5$, $C_2 = 8$ and $C_3 = 5$. Please identify the max-min fair allocation for the three flows and their bottleneck link(s).

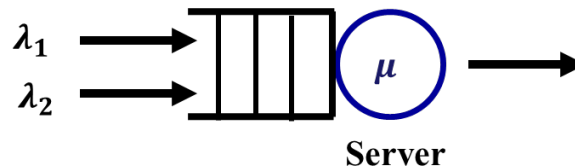
- a) Calculate the weighted max-min fair allocation $\mathbf{x} = (x_1, x_2, x_3)$ to the three flows when the weights satisfy $\phi_1 : \phi_2 : \phi_3 = 2 : 1 : 1$.
- b) Calculate the weighted max-min fair allocation $\mathbf{x} = (x_1, x_2, x_3)$ to the three flows when the weights satisfy $\phi_1, \phi_2, \phi_3 = 1 : 2 : 2$.

Problem 3

Under an M/M/1 system that uses First-In-First-Out (FIFO) scheduling policy. Suppose the arrival rate is λ and the service rate is μ .

- a) If a packet arrives and finds that the queue is empty, but the server is busy, what is this packet's mean sojourn time?
- b) If a packet arrives and finds that there are n other packets in the queue, what is this packet's mean sojourn time?
- c) Can you derive the mean sojourn time for a typical packet?

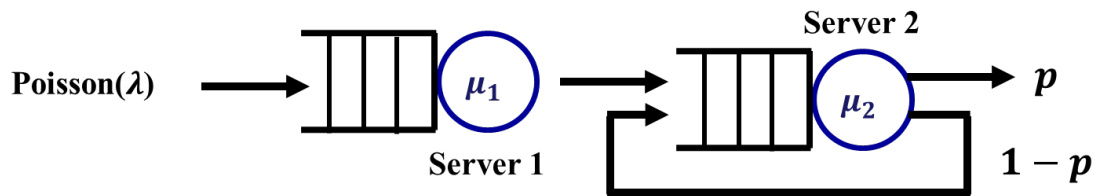
Problem 4



Two packet flows, one TCP with arrival rate $\lambda_1 = 200$ per second and one UDP with arrival rate $\lambda_2 = 500$ per second, go through a common link, which can be modelled as a single-server queueing system with a queue of infinity length. For the TCP packet flow, half of the packets have a length of 50 bytes and another half of the packets have a length of 100 bytes. All UDP packets have the same length of 30 bytes. The link has a processing rate of $5 * 10^4$ bytes per second.

- a) What is the utilization of the link?
- b) If the average number of packets in the system is 1000, what is the average queueing delay for these packets?

Problem 5



Consider the tandem queue structure where the arrival is Poisson with rate λ and the service times are exponential with rate μ_1 and μ_2 . After getting served by Server 2, each job has a probability of p to leave the system and a probability of $1 - p$ to join the queue of Server 2 again. Suppose that the system has run a long time and is stable.

- When you take a random observation of the system, what is the probability that both queues of the system are empty?
- When you take a random observation of the system, what is the probability that both queues have the same number of jobs?